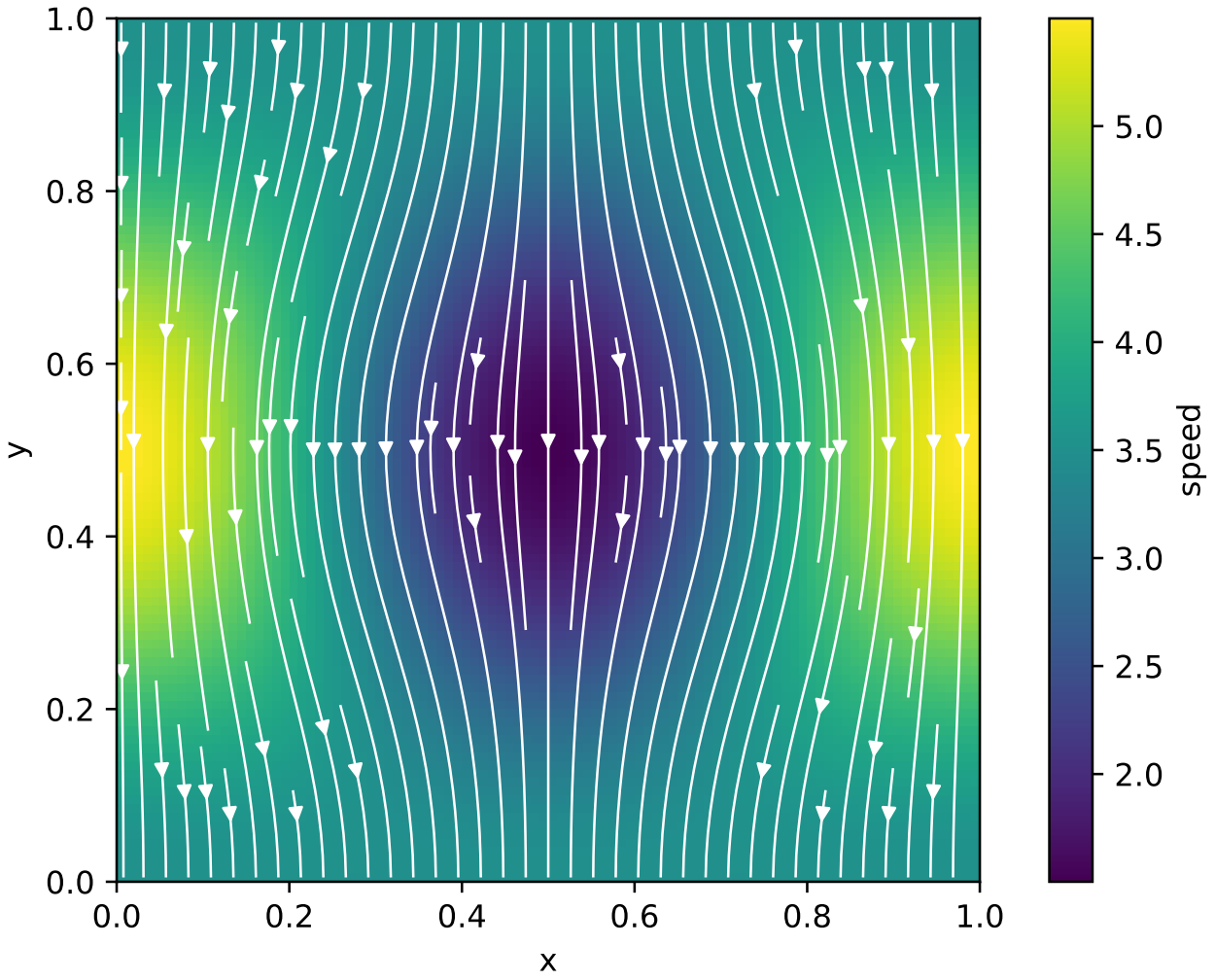
Sean Gravelle
Numerical PDEs Homework #1

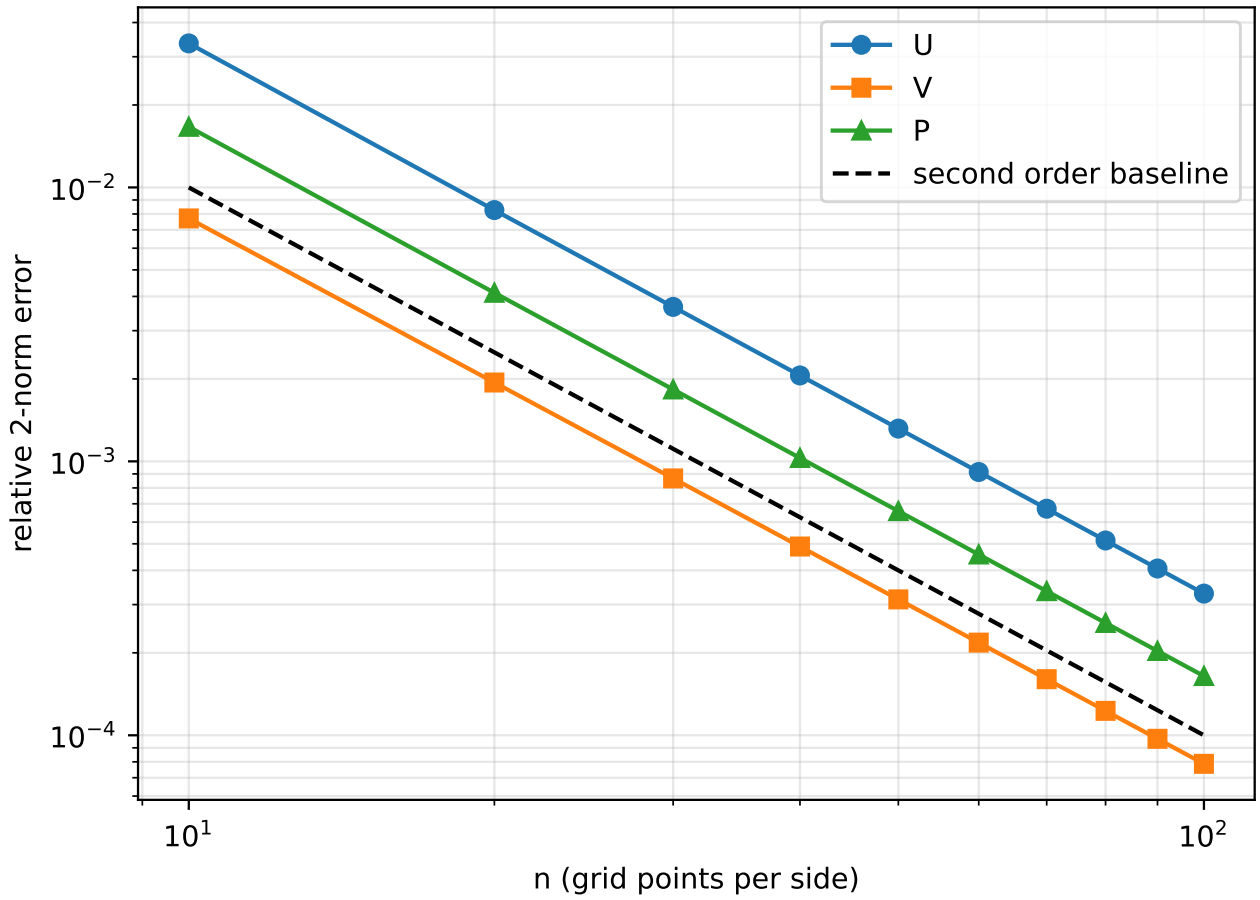
Problem 1: Solve Stokes

My solution to the Stokes equations in 2 dimensions, with the given boundary conditions, is plotted below. I used $n = 100 = Nx = Ny$ gridpoints in both x and y directions. Arrows indicate direction of the fluid's flow and color indicates the magnitude of the solution vector; that is, the speed of the fluid.



Problem 2: Error Analysis

Plotting the relative error of numerical solutions on an $n \times n$ grid gives the following errors for U, V, and P. The slopes of each show that their errors decrease proportionally to $1/n^2$ (the second order baseline plotted as a black dashed line).



Problem 3: Changing the boundary conditions

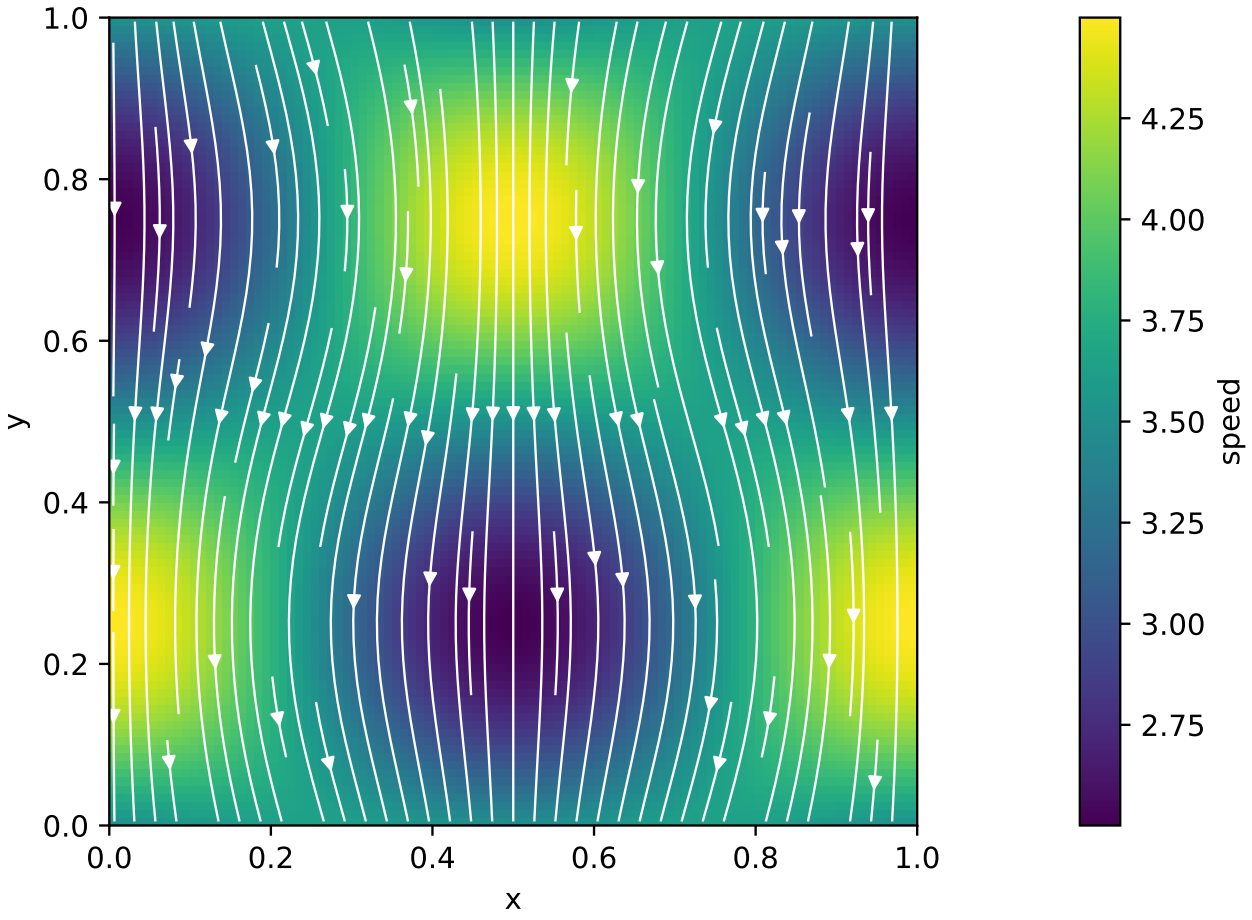
I modified my code to solve the same PDE but with different boundary conditions on U:

- On the left and right boundaries: Dirichlet condition $U = 0$
- On the top and bottom boundaries: Neumann conditions $\partial U/\partial y = 0$

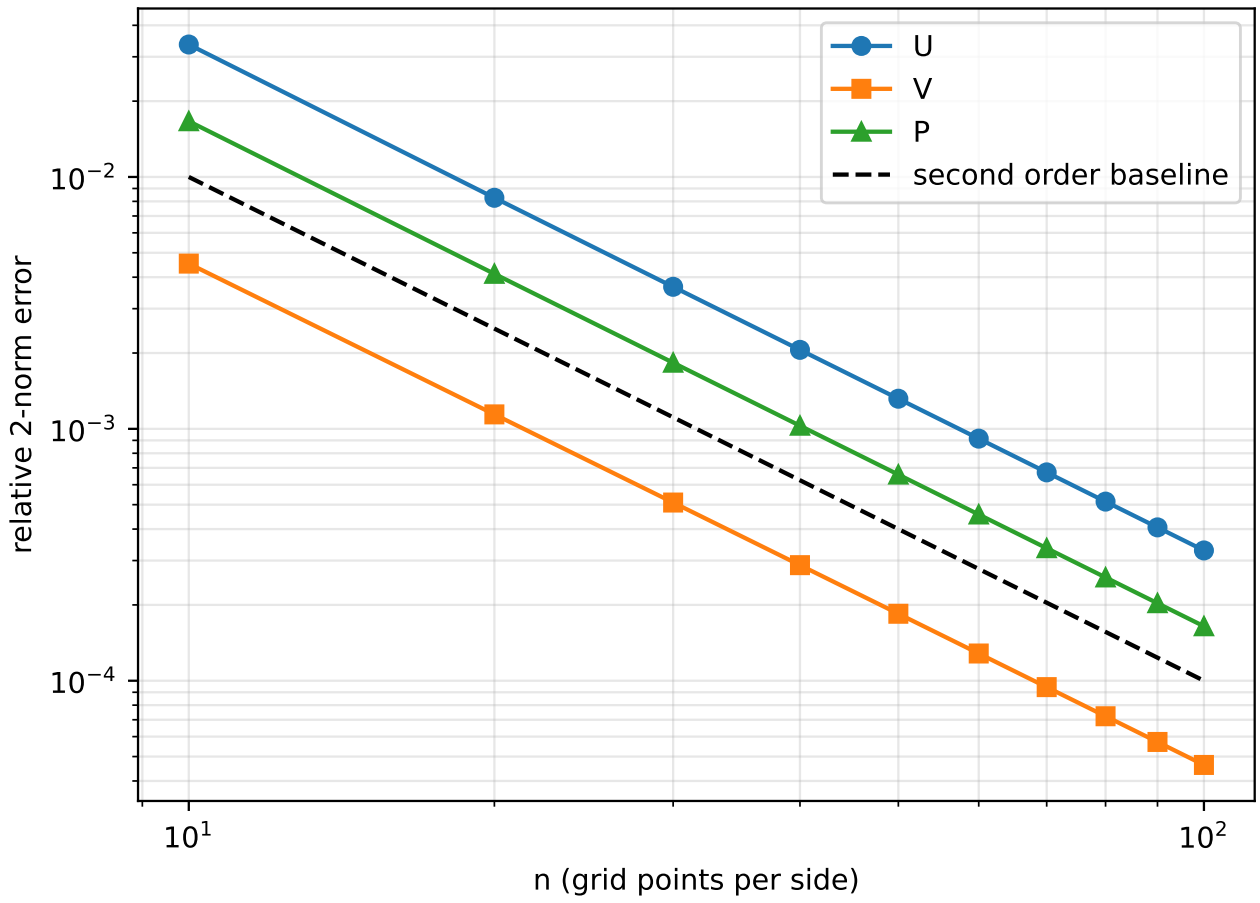
I kept the same boundary conditions on V and P as in the original problem. Using the method of manufactured solutions, my analytic functions matching these boundary conditions are:

$$
\begin{aligned}
U_{an}(x, y) &= \cos(2\pi y)\sin(2\pi x) \\
V_{an}(x, y) &= -3.5 - \cos(2\pi x)\sin(2\pi y) \\
P_{an}(x, y) &= \sin(2\pi y)\cos(2\pi x)
\end{aligned}
$$

My solution with these new boundary conditions is shown below.



The plot below shows the error of my solution to the problem with modified boundary conditions. Error is measured in the 2-norm, as with the original problem. Second order accuracy is once again obtained.



AI disclosure statement: All plots were made with LLM assistance. All mathematical work and all code to produce the numerical solutions and error calculations are my own, without LLM.